

WIPRO NGA Program _26SUB0726-JAVA J2EE INTEGRATION

Capstone Project Presentation – 23rd June 2026

Project Title Here –Courier Tracking System

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Courier Tracking System

Introduction

- The Courier Tracking System is a web-based application developed to automate courier operations and provide real-time shipment tracking.
- The application helps manage customers, parcels, shipments, tracking information, and delivery details efficiently.
- The system eliminates manual record keeping, improves shipment visibility, reduces operational overhead, and ensures secure and reliable courier management.

Proposed System

The proposed Courier Tracking System automates the complete courier lifecycle..

The system provides:

- Customer Registration and Management
- Parcel Management
- Shipment Creation and Status Updates
- Real-Time Tracking
- Delivery Management
- Role-Based Access
- Control Secure API Access
- Dashboard Monitoring
- The solution is scalable, secure, and userfriendly.

Project Objectives

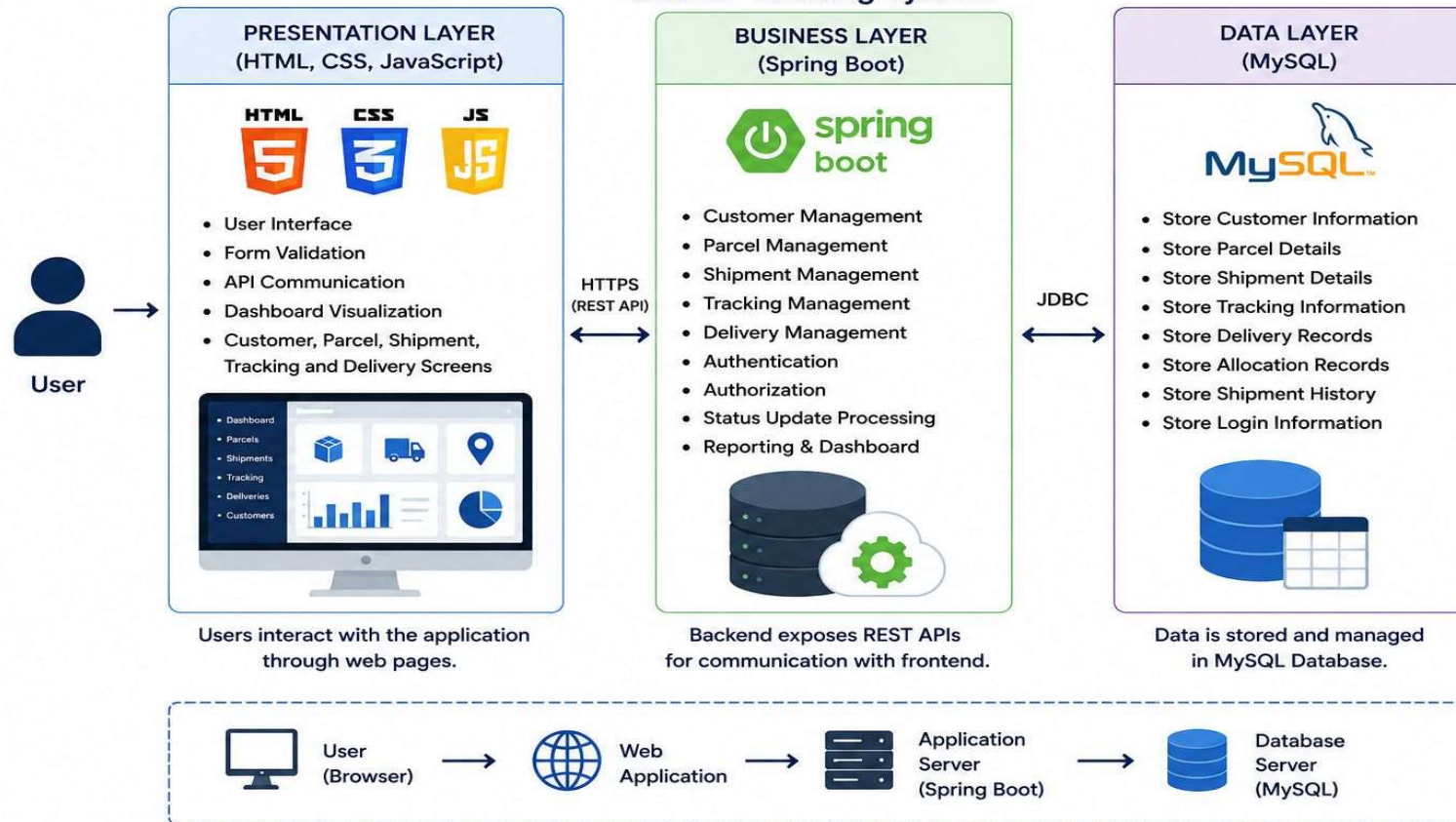
Objectives

- ❑ Automate courier management processes.
- ❑ Manage customers and parcels efficiently.
- ❑ Provide real-time shipment tracking.
- ❑ Maintain complete shipment history.
- ❑ Secure application using Spring Security.
- ❑ Monitor application health using Actuator.
- ❑ Deploy application using Docker and Kubernetes.
- ❑ Automate build and deployment using Jenkins CI/CD.

System Architecture

THREE – TIER ARCHITECTURE

Courier Tracking System



Presentation Layer

The project follows a three-tier architecture.

Developed Using

- HTML
- CSS
- JavaScript
- Bootstrap

Responsibilities

- User Interface
- Form Validation
- Dashboard Display
- API Communication
- Customer Management Screens ,Parcel Screens ,Shipment Screens
Tracking Screens

Business Layer

Developed Using

- Spring Boot

Responsibilities

- Customer Management
- Parcel Processing
- Shipment Creation
- Tracking Updates
- Delivery Processing
- Authentication and Authorization
- Exception Handling
- Validation

The backend exposes REST APIs for communication with the frontend.

Data Layer

Developed Using

- MySQL Database

Responsibilities

- Store Customer Information
- Store Parcel Details
- Store Shipment Details
- Store Tracking Information
- Store Delivery Records
- Maintain Shipment History

Data persistence is implemented using Spring Data JPA and Hibernate.

System Modules & Features

The Courier Tracking System consists of two primary user roles:

Administrator

- ❑ Manage Customers
- ❑ Manage Parcels
- ❑ Create Shipments
- ❑ Update Shipment Status
- ❑ Manage Tracking Information
- ❑ Manage Deliveries View Dashboard

Customer

- ❑ Login Using Phone Number and Parcel ID
- ❑ View Parcel Details
- ❑ Track Shipment Status
- ❑ View Delivery Information

System Workflow

Admin Login



Customer Registration



Parcel Creation



Shipment Creation



Tracking Number Generation



Shipment Status Updates



Delivery Processing



General - RPS
Data

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DevOps, Deployment & Benefits

- The Courier Tracking System is developed using modern technologies including HTML,CSS Java Script for the frontend, Spring Boot for the backend, and MySQL for data storage. Security is implemented using Spring Security and , while Swagger is used for API documentation and Spring Boot Actuator for monitoring application health and metrics.
- To ensure portability and consistency across environments, the application is containerized using Docker. Kubernetes is used to orchestrate and manage the deployment of application containers, providing reliability and simplified deployment management. A Jenkins CI/CD pipeline automates the software delivery process by performing source code checkout, application build, testing, Docker image creation, and Kubernetes deployment.

Testing and Validation

Controller Layer Testing

- Customer APIs
- Parcel APIs
- Shipment APIs
- Tracking APIs
- Delivery APIs

Service Layer Testing

- Business Logic Validation
- Repository Mocking using Mockito

Benefits

- ☐ Automated courier management
- ☐ Real-time shipment tracking
- ☐ Secure authentication
- ☐ Efficient parcel handling
- ☐ Centralized data management
- ☐ Docker and Kubernetes deployment
- ☐ Jenkins CI/CD automation
- ☐ Improved operational efficiency

☐ Drawbacks:

- ☐ Requires regular maintenance
- ☐ Increased infrastructure resources
- ☐ Single database dependency
- ☐ Limited analytics and reporting
- ☐ No GPS integration in current version

Conclusion

- The Courier Tracking System successfully automates the complete courier lifecycle including customer management, parcel registration, shipment creation, tracking, and delivery management.
- The project improves operational efficiency through centralized management, secure access control, automated workflows, and modern DevOps practices using Docker, Kubernetes, and Jenkins.

ER Diagram

